animATEM user guide

animATEM is an ATEM switcher control app that shows you a SuperSource or DVE layout before you commit it. Alongside ordinary cut/auto/FTB switching, it captures the switcher's own multiview over USB, crops the source boxes out of that live feed and recomposites them into your arrangement, so you're dragging real pixels around rather than guessing at coordinates.

> **Before you rely on this:** everything here is built and exercised (typecheck, lint and tests pass, the control server round-trips real commands, the packaged app boots), and the capture and calibration path **has been run against a real ATEM Mini Extreme ISO** (2026-07-17/18), which is where a wrong guess about multiview window numbering turned up and got fixed. **The cut, auto, FTB and recall commands have only run in harnesses, never against a real switcher.** Read What isn't proven yet before you put this anywhere near an audience.
>
> This is an AI-assisted codebase, human-directed and reviewed. Review it before relying on it.

What isn't proven yet

Two things to hold in your head, because they pull in opposite directions:

1. **The Preview panes' scale is a labelled placeholder.** The numbers ATEM uses for SuperSource and DVE box position and size are raw wire values, and their real-world scale isn't documented anywhere reachable without hardware. animATEM guesses. **A box may be drawn at the wrong size or position relative to what the switcher will actually do.**
2. **What gets pushed is not affected by that guess.** Take sends the exact raw values you set, untouched. A wrong preview scale makes the *picture* wrong, not the *command*.

So treat the preview as a rough guide, not proof, until you've calibrated against real output. Don't put this in front of an audience on its first outing.

What you need

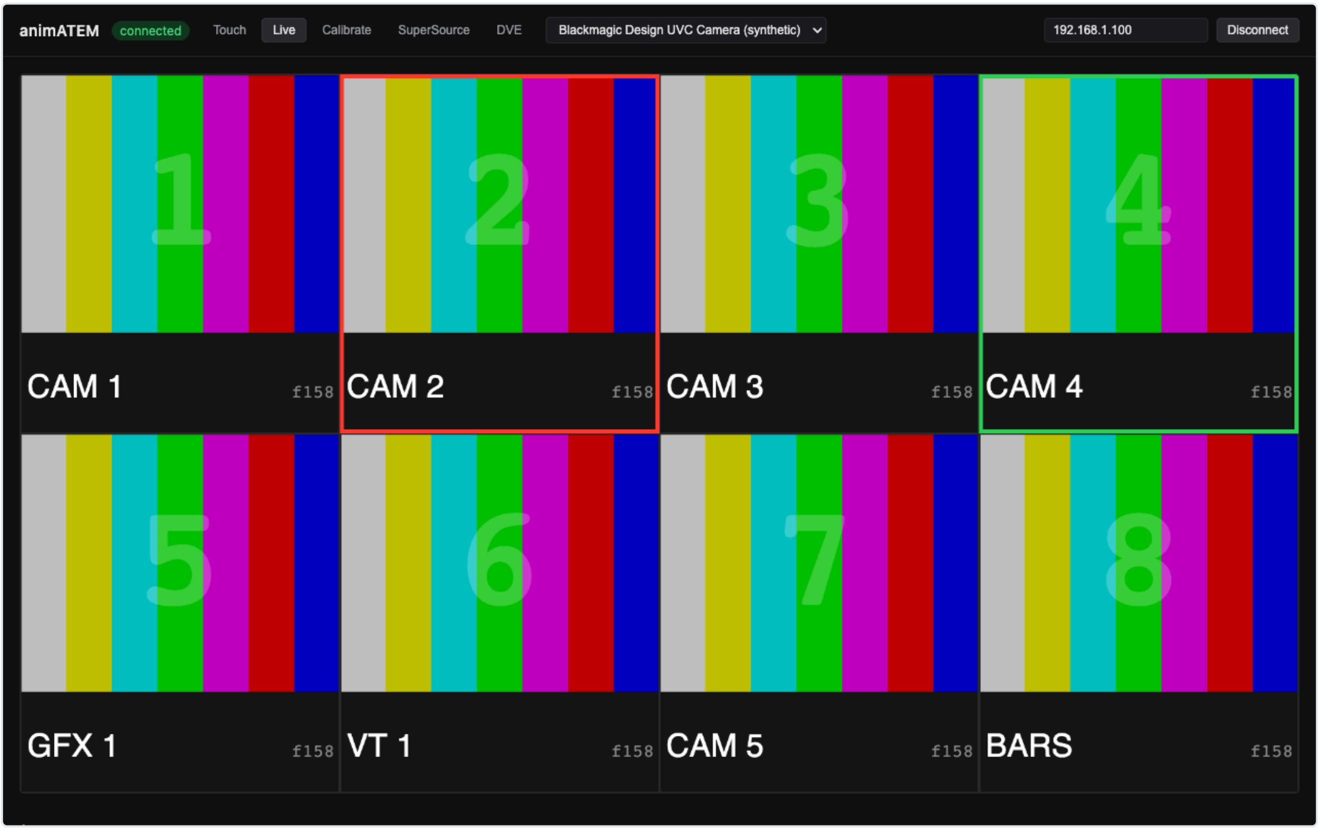
- An **ATEM Mini Pro or Extreme ISO** (the Phase 1 family).
- **The switcher's USB output set to Multiview**, not the default Program. The compositing workflow reads the multiview feed. With USB set to Program you get one picture and none of the source boxes, and nothing works as it should.
- A network route to the switcher for control, and the USB cable for capture.

The two halves

It's worth keeping these separate in your head, because only one of them is unusual.

Standard switching: cut, auto, fade-to-black, program/preview selection, aux routing. Plain ATEM control over the network, and it behaves the way any other ATEM controller would.

SuperSource and DVE layout compositing is the unusual part. Rather than sending a layout command and hoping, animATEM captures the switcher's **own multiview output over USB** (it enumerates as a UVC webcam), crops the individual source boxes out of that live feed, and recomposites them into your arrangement **with real live pixels**. You then **Take** it to push it to the switcher, or **Animate** it (see Animating a SuperSource move).

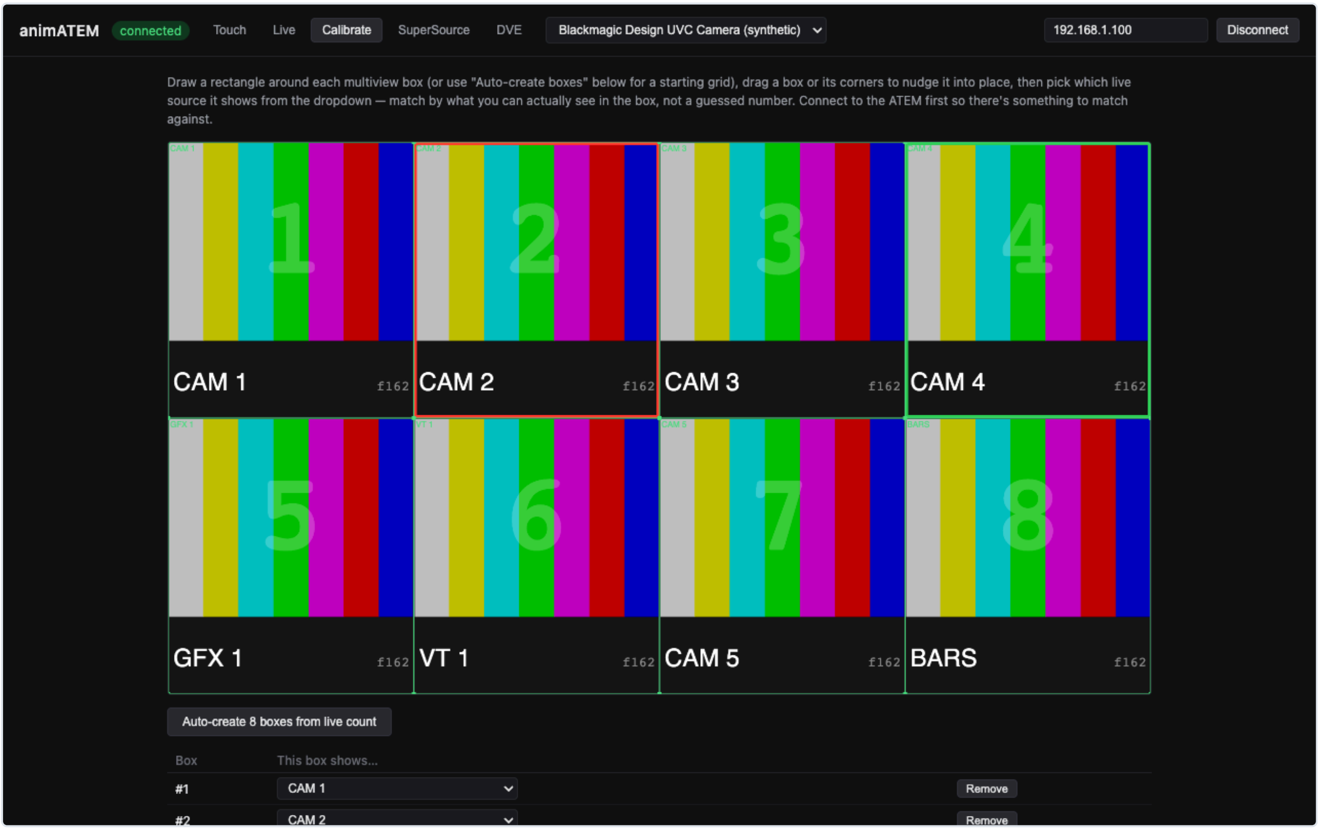

animATEM
connected
Touch
Live
Calibrate
SuperSource
DVE
Blackmagic Design UVC Camera (synthetic)
192.168.1.100
Disconnect
1
CAM 1
f158
2
CAM 2
f158
3
CAM 3
f158
4
CAM 4
f158
5
GFX 1
f158
6
VT 1
f158
7
CAM 5
f158
8
BARS
f158

*The Live tab. **Every screenshot in this guide is the app's own synthetic test feed, not a real ATEM**: the source selector reads "Blackmagic Design UVC Camera (synthetic)" in all of them.*

Calibrate first

The capture path needs aligning to the actual video before the composite means anything. On the **Calibrate** tab you draw a rectangle around each multiview box (or press **Auto-create boxes** for a starting grid), nudge the corners into place, and tell the app which live source each box is showing.

Match by **what you can actually see in the box**, not by a guessed number. Connect to the ATEM first, so there's something to match against.


animATEM
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Blackmagic Design UVC Camera (synthetic)
192.168.1.100
Disconnect
1
2
3
4
CAM 1
f162
CAM 2
f162
CAM 3
f162
CAM 4
f162
5
6
7
8
GFX 1
f162
VT 1
f162
CAM 5
f162
BARS
f162
Auto-create 8 boxes from live count
Remove
Remove

Draw a rectangle around each multiview box (or use "Auto-create boxes" below for a starting grid), drag a box or its corners to nudge it into place, then pick which live source it shows from the dropdown — match by what you can actually see in the box, not a guessed number. Connect to the ATEM first so there's something to match against.

Box	This box shows...
#1	CAM 1
#2	CAM 2

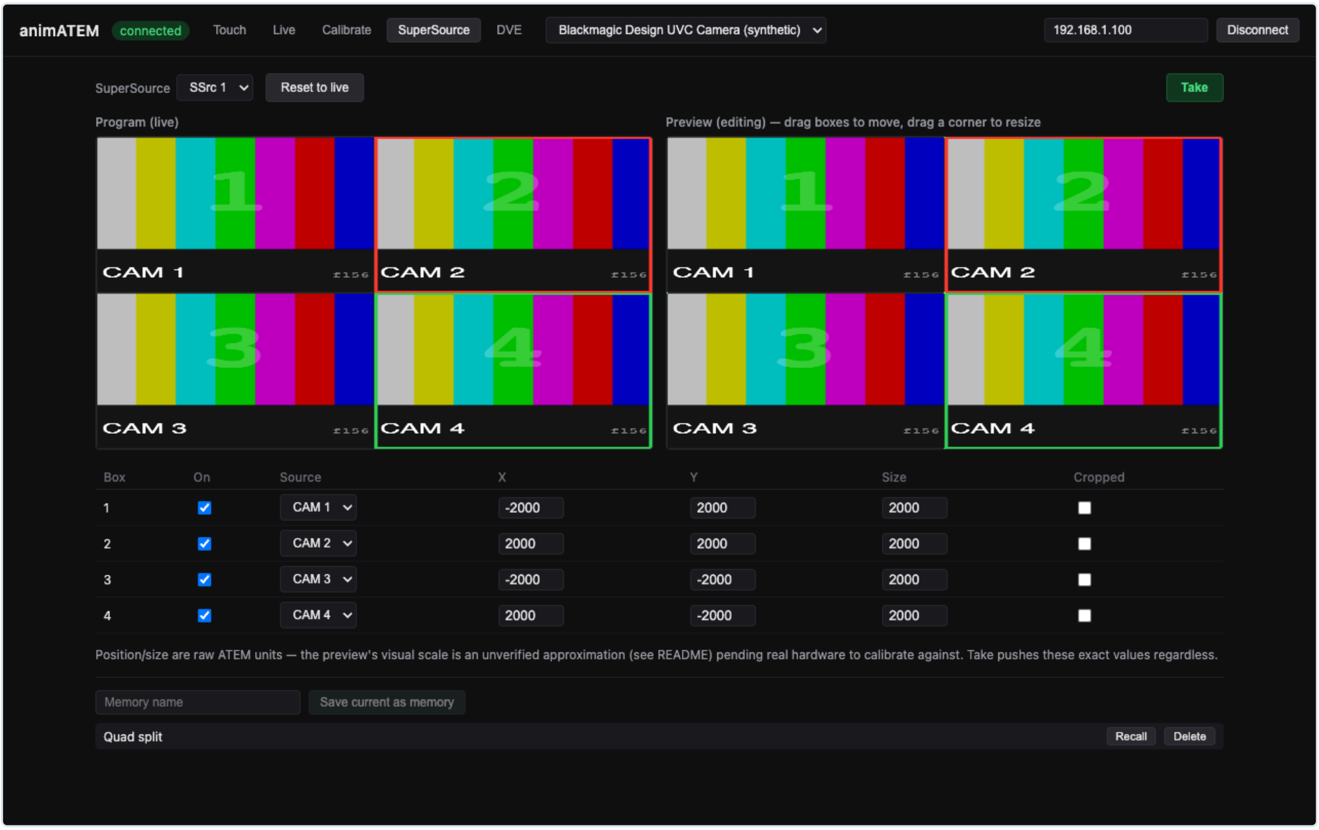
Two things about calibration that will bite otherwise:

- **A profile is stored per resolution.** Change the multiview resolution and you need to calibrate again. The app won't reuse the old profile, and it won't warn you that you're running uncalibrated.
- **An uncalibrated composite looks plausible and is wrong**, which is the worst failure mode available. Don't assume it's pixel-true until you've done this.

Preview, Take and what's editable

The editor works Preview → Take, like the switcher itself. The left pane is what's live; the right pane is what you're building.

- Drag a box to move it, drag a corner to resize.
- **SuperSource boxes scale uniformly**: one size value, so they stay square-proportioned.
- **DVE boxes have independent width and height.**
- Crops (SuperSource) and masks (DVE) are separate settings, and are only applied when their enable flag is on.
- **Take** pushes the layout to the switcher.


animATEM
connected
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Live
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SuperSource
DVE
Blackmagic Design UVC Camera (synthetic)
192.168.1.100
Disconnect
Reset to live
Take
CAM 1
CAM 2
CAM 3
CAM 4
Save current as memory
Recall
Delete

SuperSource: SSrc 1

Program (live)

Preview (editing) — drag boxes to move, drag a corner to resize

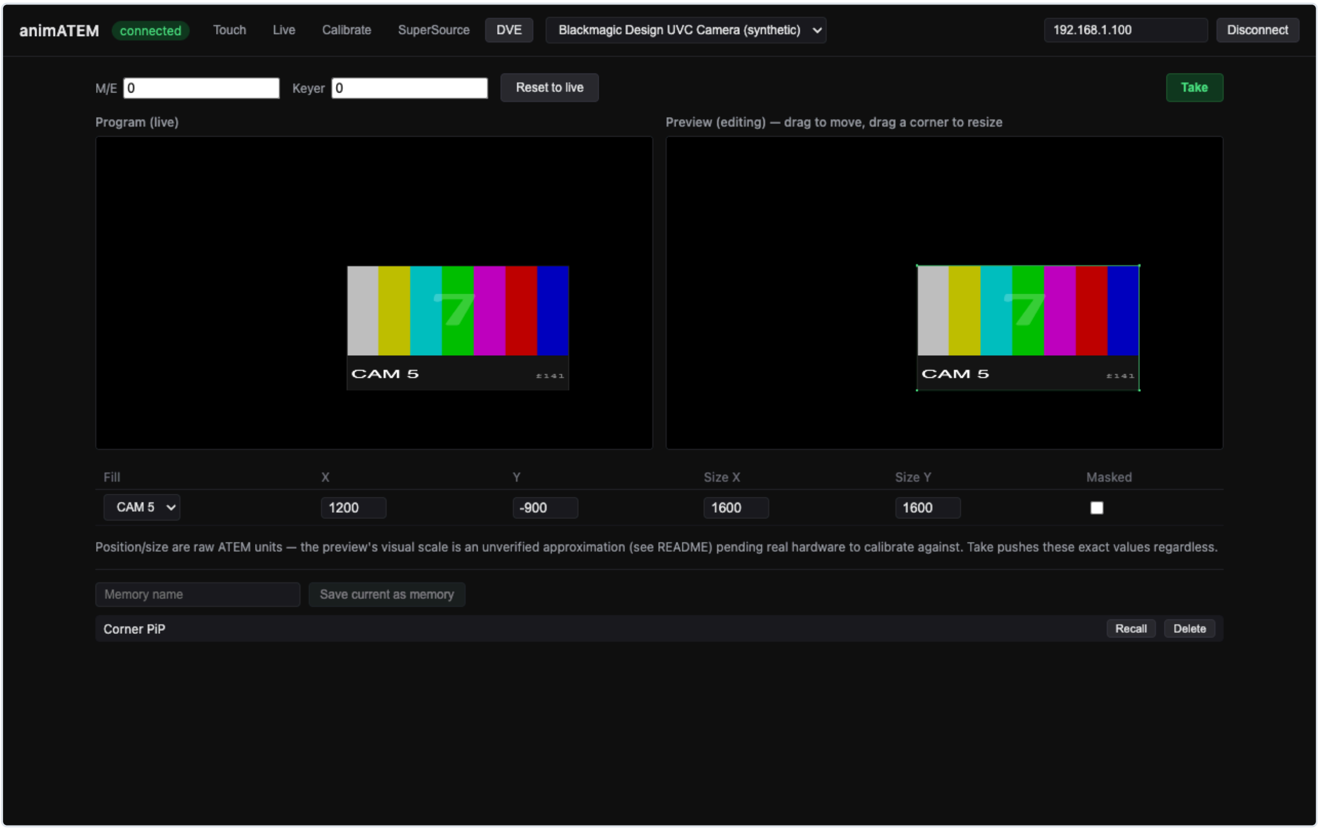
Box	On	Source	X	Y	Size	Cropped
1	☑	CAM 1	-2000	2000	2000	☐
2	☑	CAM 2	2000	2000	2000	☐
3	☑	CAM 3	-2000	-2000	2000	☐
4	☑	CAM 4	2000	-2000	2000	☐

Position/size are raw ATEM units — the preview's visual scale is an unverified approximation (see README) pending real hardware to calibrate against. Take pushes these exact values regardless.

Memory name

Quad split

The X/Y/Size fields under the panes are the raw ATEM units being sent. They're the values Take pushes verbatim, which is why they're on show rather than hidden behind the drag handles.


animATEM
connected
Touch
Live
Calibrate
SuperSource
DVE
Blackmagic Design UVC Camera (synthetic)
192.168.1.100
Disconnect
Reset to live
Take
CAM 5
Save current as memory
Recall
Delete

M/E: 0
Keyer: 0

Program (live)

Preview (editing) — drag to move, drag a corner to resize

Fill	X	Y	Size X	Size Y	Masked
CAM 5	1200	-900	1600	1600	☐

Position/size are raw ATEM units — the preview's visual scale is an unverified approximation (see README) pending real hardware to calibrate against. Take pushes these exact values regardless.

Memory name

Corner PiP

Animating a SuperSource move

The ATEM protocol has no native SuperSource tweening. An upstream keyer's DVE is animated by the switcher itself, in hardware, via fly-keyframes. SuperSource isn't.

So animATEM does it client-side: pick a duration and press **Animate** instead of **Take**, and it eases into the target layout by streaming a fixed cadence of interpolated positions to the switcher.

Two things follow from that, and both matter on a show:

- **It's at the mercy of the network and the app.** A hardware DVE fly is one command the switcher executes on its own; a SuperSource animation is a continuous stream from your laptop. A network hiccup, a busy machine, or the app being backgrounded all show up in the move.
- **There's no DVE equivalent of Animate**, because there doesn't need to be. The switcher already does that one properly, in hardware.

Memories

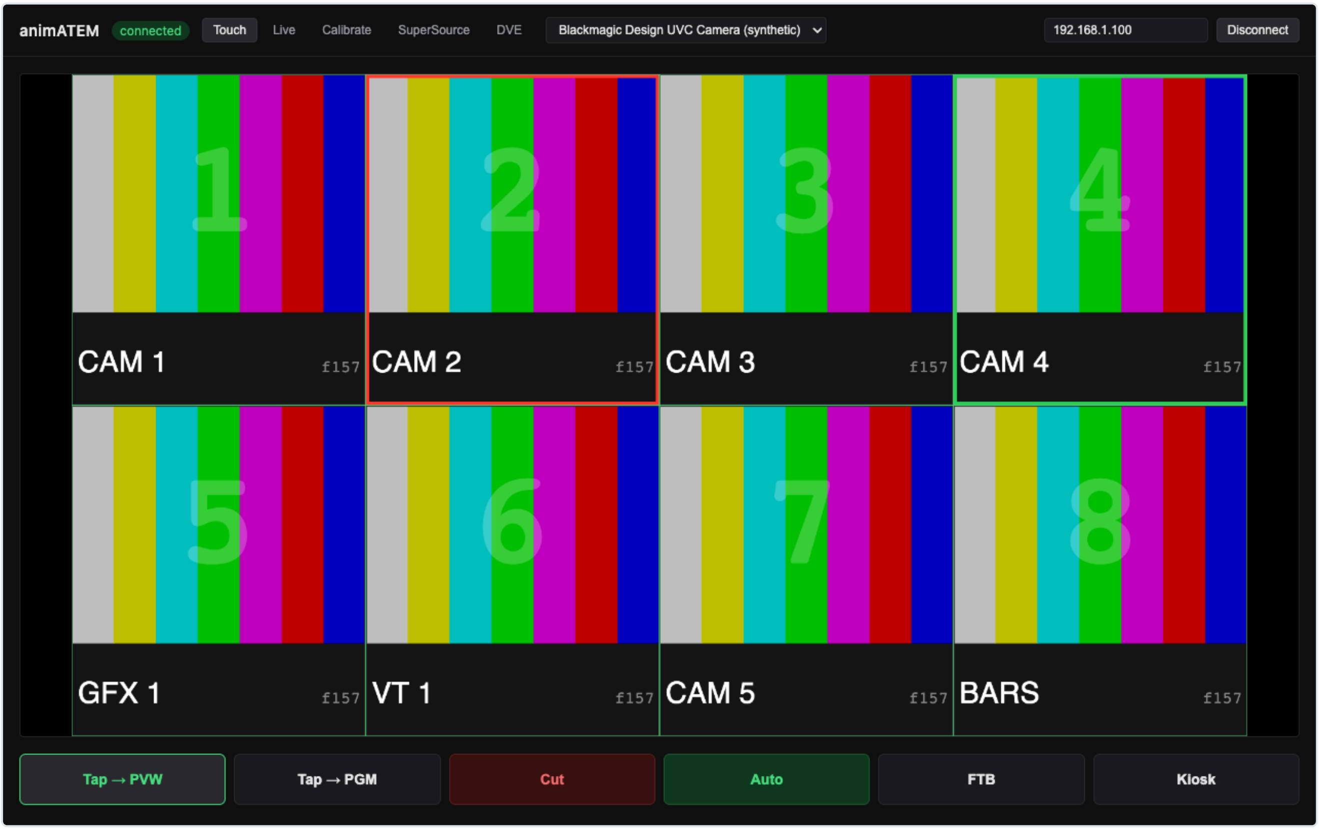
Named presets that capture a SuperSource or DVE arrangement. They're **animATEM's own**, stored in a JSON file in the app's data directory, and **entirely independent of the ATEM's own macro system**. They don't appear on the switcher, and they won't survive a move to a different machine unless you copy the file.

- Recall from **the app's own UI** loads the memory into an **editable Preview**. Nothing goes to air until you Take.
- Recall from **a Stream Deck / Companion button goes straight to air** (see Stream Deck and Companion).
- Saving a memory with an existing name or id **replaces** it.
- **If the memories file is unreadable or corrupted, the app shows an empty list** rather than an error. "All my memories vanished" usually means a damaged file, not a deletion.

Back the file up before a show if the memories matter.

The touch layout

A separate operator view sized for a touchscreen: tap a source to send it to preview or program, with Cut, Auto, FTB and a kiosk toggle on a bottom bar.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f157
4
CAM 4
f157
5
GFX 1
f157
6
VT 1
f157
7
CAM 5
f157
8
BARS
f157
Tap → PVW
Tap → PGM
Cut
Auto
FTB
Kiosk

Tap → PVW and **Tap → PGM** decide what a tap on a source does. Leaving it on **Tap → PGM** means every tap is a live cut, which is the point on a fast show and a hazard on a slow one.

Stream Deck and Companion

A Bitfocus Companion module lives in its own repo, companion-module-animatem. Buttons can trigger Cut, Auto, FTB, source selection, memory recall and animated SuperSource recall, with feedback and variables for the current program and preview input.

> **A Companion memory-recall button goes straight to air.** That's on purpose: a physical button press is treated as a physical action, the way a memory bank works on real switcher hardware. But it's the **opposite** of what the same recall does in the app's own UI, which loads into an editable Preview. **Label those buttons so nobody expects a preview.**

Other behaviours worth knowing before you build a page of buttons:

- **Commands are never acknowledged.** The module learns whether something worked by watching the next state update, not from a reply. A command sent to a disconnected switcher looks exactly like one that worked.
- **An unknown memory id does nothing, silently.**
- **Asking to animate a DVE memory does nothing, silently**, because animation is SuperSource-only.

- The memory list is pushed on connect and when the app broadcasts it, so a surface can hold a stale list if memories change underneath it.

Security

The control server binds to `127.0.0.1:51234` with **no authentication**. Anything that can reach that port **on the local machine** can cut, auto, FTB or recall a memory, which means putting something on air.

That's only acceptable because it's localhost-only, which is the default and the only supported configuration. **Don't expose it to the network.** It has no authentication to add safety if you do.

Troubleshooting

Symptom	Cause
No source boxes in the capture, just one picture	The switcher's USB output is set to Program. It must be **Multiview**.
Composite is offset or the wrong scale	Not calibrated, or calibrated at a different resolution — profiles are per-resolution.
Preview doesn't match what the switcher did	Expected for now. The preview coordinate scale is an unverified placeholder; the *push* is exact.
A SuperSource animation stutters	It is streamed from this machine, not run in the switcher. Network or CPU.
DVE has no Animate button	Correct — the switcher animates DVEs itself, in hardware.
A Companion button put something on air unexpectedly	Memory recall from Companion is immediate by design.
A Companion button appears to do nothing	Commands aren't acknowledged; the switcher may be disconnected. Watch the state feedback, not the button.
All memories disappeared	Usually a corrupted memories file — it fails to an empty list rather than erroring.
Memories missing on another machine	They are stored per-machine in the app's data directory, not on the switcher.
macOS says the app is damaged	Not the released build — those are signed and notarised. A self-built or pre-notarisation copy is quarantined; see UNSIGNED.md.

See also

- API.md: the control protocol, state model and coordinate conversions
- DEVELOPING.md: building and extending
- companion-module-animatem: the Companion module, in its own repo
- README: concept, status and the security note

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